The two–row d=4 fiber law as a continued fraction: a Chebyshev / recessive–solution reformulation

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Abstract

This note records a structural reformulation of the two–row d=4 fiber–vanishing theorem $G_{(a,b)}(i) = 0 \iff (a, b) = (2, 2)$. Three things are proved (all verified exactly). *(i)* The relevant coefficients are Chebyshev/Fourier coefficients of a pure power: $\phi_j := r_{m-j} = \frac{1}{2\pi}\int_0^{2\pi}(1 + i + 2\cos\theta)^m e^{ij\theta}\, d\theta$, i.e. the j-th Chebyshev coefficient of $(2c+1+i)^m = 2^m(c-c_0)^m$ with $c_0 = -\frac{1+i}{2}$. *(ii)* The whole two–row law is the single "clockwise–turning" statement $\mathrm{Im}(\overline{\phi_j}\phi_{j+1}) < 0$ for $1 \le j \le m-1$; a sign recurrence shows this is *irreducible to magnitude information* (sign data alone is self–referential), which is exactly why a genuine magnitude bound (the historical "Lemma 1") was needed. *(iii)* ϕ_j is the *recessive* (minimal) solution of the three–term recurrence, so by Pincherle's theorem the ratio $\eta_j = \phi_{j+1}/\phi_j$ is a terminating continued fraction; with $\xi_j = (m + j + 1)\eta_j$,

$$\xi_{j-1} = \frac{(m+j)(m-j+1)}{(1+i)\, j + \xi_j}, \qquad \xi_m = 0,$$

and *the law is precisely that every tail ξ_j lies in the open lower half–plane.* This recasts the problem in the classical theory of continued–fraction value regions, the natural home for "all tails in a half–plane" statements, and supersedes the axis–aligned envelope/box programme, which I show is structurally inadequate (the fixed–j asymptotics it relies on are invalid for $j \gtrsim \sqrt{m}$, while the band runs to $j \sim 2m/3$). The remaining gap is reduced to one clean classical question: exhibit a value region $V \subseteq \{\mathrm{Im}\, w < 0\}$ invariant under the maps $s_n(w) = (m+n)(m-n+1)/((1+i)n + w)$.

1 Setup and the engine

Fix $m \ge 4$, $f(u) = u^2 + (1+i)u + 1$, $r_l = [u^l]f(u)^m \in \mathbb{Z}[i]$, $N_l = |r_l|^2$. The proved reduction (memory: `gapB-phase-positivity`, `tworow-d4-Nratio`) gives: $G_{(a,b)}(i) = r_b - r_{b-1}$, and the two–row law $G_{(a,b)}(i) = 0 \iff (a, b) = (2, 2)$ is equivalent to $Q_l := \mathrm{Im}(r_l\overline{r_{l-1}}) > 0$ for $1 \le l \le m-1$. The Baxterised recurrence (verified in $\mathbb{Z}[i]$) is

$$(l + 1)\, r_{l+1} = (1 + i)(m - l)\, r_l + (2m - l + 1)\, r_{l-1}. \tag{1}$$

2 Chebyshev / Fourier reformulation

Lemma 1. *On the unit circle $f(e^{i\theta}) = e^{i\theta}\,(1 + i + 2\cos\theta)$, hence $|f(e^{i\theta})|^2 = (1 + 2\cos\theta)^2 + 1$. Writing $\phi_j := r_{m-j}$ (well defined for all $j \in \mathbb{Z}$ since f is reciprocal, $r_l = r_{2m-l}$, so $\phi_{-j} = \phi_j$),*

$$\phi_j = \frac{1}{2\pi}\int_0^{2\pi}\big(1 + i + 2\cos\theta\big)^m e^{ij\theta}\, d\theta = \frac{1}{2\pi}\int_0^{2\pi}\big((1 + 2\cos\theta) + i\big)^m e^{ij\theta}\, d\theta. \tag{2}$$

Equivalently ϕ_j is (up to the standard normalisation) the j-th Chebyshev coefficient of the degree–m polynomial $(2c+1+i)^m = 2^m(c-c_0)^m$, $c_0 = -\frac{1+i}{2}$, expanded on $[-1,1]$ with $c = \cos\theta$.

Proof. $e^{2i\theta}+1 = 2e^{i\theta}\cos\theta$, so $f(e^{i\theta}) = e^{2i\theta}+(1+i)e^{i\theta}+1 = e^{i\theta}(1+i+2\cos\theta)$; taking $|\cdot|^2$ gives the modulus formula. Since $f(u)^m = \sum_l r_l u^l$, on $|u|=1$ we have $f(e^{i\theta})^m = \sum_l r_l e^{il\theta}$, and $f(e^{i\theta})^m = e^{im\theta}(1+i+2\cos\theta)^m$; multiplying by $e^{-il\theta}$ and integrating, $r_l = \frac{1}{2\pi}\int e^{i(m-l)\theta}(1+i+2\cos\theta)^m d\theta$, which is (2) with $j = m-l$. Finally $1+i+2\cos\theta = 2(\cos\theta - c_0)$ with $c_0 = -(1+i)/2$, and $\cos j\theta = T_j(\cos\theta)$ identifies the Chebyshev coefficient. $\square$

In these coordinates $N_l = |\phi_j|^2$ $(j = m-l)$, the consecutive–ratio is $\mu_l = N_l/N_{l-1} = |\phi_j|^2/|\phi_{j+1}|^2$, and $\zeta_l = r_{l-1}/r_l = \phi_{j+1}/\phi_j$. Dividing (1) and relabelling gives the clean *outward* recurrence (verified numerically for $m \le 14$ and algebraically equivalent to (1)):

$$(m+j+1)\,\phi_{j+1} = (m-j+1)\,\phi_{j-1} - (1+i)\,j\,\phi_j. \tag{3}$$

3 The law is a clockwise turning, and is irreducible to magnitudes

Set $\eta_j := \phi_{j+1}/\phi_j$ and $X_j := \operatorname{Im}(\overline{\phi_j}\phi_{j+1}) = |\phi_j|^2 \operatorname{Im}\eta_j$. Since $\phi_{j+1}\overline{\phi_j} = r_{l-1}\overline{r_l} = P_l - iQ_l$ (with $l = m-j$), we have $X_j = -Q_l$. Hence:

Proposition 1. *The two–row law is exactly $X_j < 0$ (equivalently $\operatorname{Im}\eta_j < 0$) for all $1 \le j \le m-1$: the planar sequence $\phi_0, \phi_1, \ldots, \phi_{m-1}$ turns strictly clockwise.*

Proposition 2 (Sign recurrence; irreducibility)**.** *Taking* Im *of* (3) *multiplied by $\overline{\phi_j}$,*

$$(m+j+1)\,X_j + (m-j+1)\,X_{j-1} = -\,j\,|\phi_j|^2 < 0 \qquad (j \ge 1). \tag{4}$$

Given $X_{j-1} < 0$, (4) yields $X_j < 0$ iff $(m-j+1)|X_{j-1}| < j|\phi_j|^2$, which unwinds exactly to "$Q_l > 0$", i.e. to the very claim. Thus the sign data is self–referential: the law cannot be propagated from sign information alone. Breaking the loop requires an external bound on the magnitude *growth of $|\phi_j|$ — precisely the role of the norm–ratio bound ("Lemma 1").*

This is, I believe, the cleanest explanation of the long–observed "Cauchy–Schwarz is circular" obstruction. Verified ($m = 14$): (4) holds and all $X_j < 0$.

4 Recessive solution and the continued fraction

The recurrence (3) (equivalently the general $(m+j+1)\phi_{j+1} = (m-j+1)\phi_{j-1} + 2c_0 j\phi_j$ for the Chebyshev coefficients of $(c-c_0)^m$) has a two–dimensional solution space. Our ϕ_j is the *recessive* (minimal) solution: the forward recurrence is numerically unstable, amplifying the dominant companion solution. By Pincherle's theorem the ratio of consecutive terms of the recessive solution is the value of a convergent continued fraction, computed by the *backward* (stable) recurrence.

Theorem 1 (Continued–fraction form of the law)**.** *Let $\xi_j := (m+j+1)\eta_j = (m+j+1)\phi_{j+1}/\phi_j$. Then*

$$\xi_{j-1} = \frac{(m+j)(m-j+1)}{(1+i)\,j + \xi_j}, \qquad \xi_m = 0, \tag{5}$$

a terminating continued fraction, and $\eta_j = \xi_j/(m+j+1)$. The two–row law is exactly

$$\operatorname{Im}\xi_j < 0 \quad \textit{for all } 1 \le j \le m-1,$$

i.e. every tail of the continued fraction (5) *lies in the open lower half–plane.*

Proof. From (3), dividing by ϕ_j, $(m+j+1)\eta_j = (m-j+1)/\eta_{j-1} - (1+i)j$, so with $\xi_j := (m+j+1)\eta_j$ we get $(m-j+1)/\eta_{j-1} = \xi_j + (1+i)j$, hence $\eta_{j-1} = (m-j+1)/\big[(1+i)j + \xi_j\big]$. Multiplying by $(m+j)$ yields $\xi_{j-1} = (m+j)\eta_{j-1} = (m+j)(m-j+1)/\big[(1+i)j+\xi_j\big]$, which is (5). The terminal value: $\phi_{m+1} = 0$ gives $\eta_m = 0$, so $\xi_m = 0$; the first backward step reproduces $\eta_{m-1} = \phi_m/\phi_{m-1} = (1-i)/(2m)$, the known seed. The equivalence of the law to $\operatorname{Im}\xi_j < 0$ is Proposition 1 together with $\operatorname{Im}\eta_j = \operatorname{Im}\xi_j/(m+j+1)$ and $m+j+1 > 0$. □

Verified exactly ($m = 20$): (5) reproduces η_j to machine precision and all $\operatorname{Im}\xi_j < 0$.

Remark 1 (A general phenomenon)**.** For any c_0 with $\operatorname{Im} c_0 < 0$ and $\operatorname{Re} c_0 \neq 0$, numerical evidence (direct quadrature, $m \leq 15$, many c_0) indicates the Chebyshev coefficients of $(\cos\theta - c_0)^m$ turn clockwise: $\operatorname{Im}(\overline{\phi_j}\phi_{j+1}) < 0$ for $j \geq 1$. The hypothesis $\operatorname{Re} c_0 \neq 0$ is necessary: at $c_0 = -i$ the turning fails. Our case $c_0 = -(1+i)/2$ sits on the locus $|c_0|^2 = \frac{1}{2}$ (where the unrequired boundary value X_0 vanishes). A proof of this general statement would prove the law for the whole family.

5 Why the value–region route, and why the box route fails

The continued fraction (5) has the standard form $K(a_n/b_n)$ with

$$a_n = (m+n)(m-n+1) > 0 \ (1 \leq n \leq m), \qquad b_n = (1+i)\,n, \quad \arg b_n = \tfrac{\pi}{4}.$$

A *value region* is a set V with $s_n(V) \subseteq V$ for all n, where $s_n(w) = a_n/(b_n + w)$; if $V \subseteq \{\operatorname{Im} w < 0\}$ and the seeds enter V, all tails lie in $\{\operatorname{Im} w < 0\}$ and the law follows. A direct computation gives, for $a_n > 0$,

$$\operatorname{Im} s_n(w) = -\,a_n\,\frac{\operatorname{Im} b_n + \operatorname{Im} w}{|b_n + w|^2} = -\,a_n\,\frac{n + \operatorname{Im} w}{|b_n+w|^2},$$

so s_n maps $\{\operatorname{Im} w > -n\}$ into the open lower half–plane. The lower half–plane itself is therefore *not* invariant (it fails where $\operatorname{Im} w < -n$); the obstruction is concentrated at the innermost, small–n elements. The remaining task is exactly to produce an invariant sub–region $V \subseteq \{\operatorname{Im} w < 0\}$ (a disk or lens), which is the classical content of the parabola / Thron–Jones value–region theorems for continued fractions with positive partial numerators and partial denominators in a fixed half–plane.

The axis–aligned envelope ("box") programme is inadequate. The earlier reduction bounded $\mu_\ell \leq U_j = (2m+3j+3)/(2m-3j)$ by propagating a rectangle $[L_j, U_j] \times [T_j, \bar{T}_j]$ in (μ_ℓ, t_ℓ) and checking four corner inequalities, with $L_j, T_j, \bar{T}_j$ taken from the fixed–j asymptotics $\mu_\ell = 1 + \frac{3(2j+1)}{2m} + \frac{36j^2+16j-1}{8m^2} + \cdots$. I tested this directly: it fails, and the failure is structural. The fixed–j expansion has effective parameter j^2/m (successive terms scale as $(j^2/m)^k$), so it is valid only for $j \lesssim \sqrt{m}$; but the band is $2 \leq j < 2m/3$, which reaches $j \sim m$. On the bulk of the band the "envelopes" are simply wrong functions. (The hard end is small j, where the slack $U_j - \mu_\ell = \frac{2j+1}{8m^2} + \cdots$ is $O(1/m^2)$ and the expansion *is* valid; the large–j end has $O(1)$ slack and is easy.) Moreover the box decorrelates (μ, t), which are tied along the orbit ($\mu - 1 \approx 6t$ to leading order); a correlated tube, not a rectangle, is required. Both defects are absent in the continued–fraction formulation, where the single complex variable ξ_j carries the correlation automatically and the value–region condition is uniform in j.

6 The remaining gap, precisely

Conjecture 1 (Value region in the lower half–plane)**.** There is a region $V \subseteq \{\operatorname{Im} w < 0\}$, e.g. a disk or half–plane depending on (m, j), with $s_n(\overline{V}\,) \subseteq V$ for the maps $s_n(w) = \dfrac{(m+n)(m-n+1)}{(1+i)n + w}$, into which the terminal data $\xi_m = 0$ feeds. Establishing this proves $\operatorname{Im} \xi_j < 0$ for all $1 \le j \le m-1$, i.e. the entire two–row d=4 law, without recourse to the $O(1/m^2)$–tight norm–ratio bound.

7 Verification

All identities verified exactly in $\mathbb{Z}[i]$/`Fraction` (scripts in `scratch/2026-06-05-lemma1/`): (2) ($m \le 12$), (3) (all j, $m \le 14$), the sign recurrence (4) with $X_j < 0$ ($m \le 14$), and the continued fraction (5) reproducing η_j to 10^{-16} with all $\operatorname{Im} \xi_j < 0$ ($m = 20$). The box–programme failure and the j^2/m expansion scale were confirmed over $6 \le m \le 3000$.

Summary of what is proved here

- **R1** (Lemma 1) $\phi_j = r_{m-j}$ are Chebyshev coefficients of the pure power $(2c + 1 + i)^m$; clean integral form (2) and outward recurrence (3).
- **R2** (Props. 1, 2) The law is the clockwise turning $X_j < 0$, and the sign recurrence (4) shows this is *irreducible* to sign data: a magnitude input is genuinely necessary.
- **R3** (Theorem 1) ϕ_j is the recessive solution; the law is "all tails of the continued fraction (5) lie in the lower half–plane."
- **R4** The axis–aligned envelope programme is structurally inadequate (invalid expansion for $j \gtrsim \sqrt{m}$; decorrelation); the value–region route is uniform and is the natural classical setting.

Remaining gap (single, classical): a lower–half–plane value region for the continued fraction (5).